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Dipartimento di
Elettronica e
Telecomunicazioni



Electronics for Embedded Systems

Department of Electronics and Telecommunications (DET)
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Experiment Title:	Lab 4 – Design and Characterization of a SAR A/D Converter
Course:	Electronics for Embedded Systems
Group:	B04
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Abstract

This laboratory focused on the design, implementation, and characterization of a Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC). The system was realized using a hybrid approach: the control logic was implemented on the Cyclone V FPGA (DE1-SoC board), the Digital-to-Analog conversion uses the on-board ADV7123 video DAC, and the comparison stage was performed by an external LM393 analog comparator mounted on a breadboard. The experiment proceeded in three phases: first, the dynamic characteristics of the LM393 comparator were measured, identifying the propagation delay as the limiting factor for the clock speed. Second, the transfer function of the DAC was verified for linearity. Finally, the complete SAR feedback loop was closed, and the system was tested by converting DC voltages and dynamic waveforms. The results demonstrated the correct functioning of the binary search algorithm, with the conversion speed effectively limited by the analog settling times.

1. Introduction

The Successive Approximation Register (SAR) ADC is a popular architecture for medium-resolution, medium-speed applications due to its efficient hardware usage. It operates on a "Divide & Conquer" technique, determining the digital output bit-by-bit from MSB to LSB. The objective of this lab was to build a custom SAR ADC by integrating digital logic with external analog components. An important aspect was understanding the timing constraints imposed by the analog loop:

$$T_{CLK} > t_{comp} + t_{logic} + t_{DAC}$$

In this specific implementation, the cycle time is dominated by the comparator delay (t_{comp}), which was measured to be approximately 330 ns. Conversely, the settling time of the video DAC (t_{DAC}) and the logic propagation delay (t_{logic}) are in the order of nanoseconds and are therefore negligible by comparison. The experiment aimed to characterize these delays to properly tune the SAR clock frequency (f_{slow}) and achieve reliable conversion.

2. Materials and Methods

Hardware:

- **FPGA Board:** Altera DE1-SoC (Cyclone V).
- **Analog Comparator:** LM393 (Open Collector output).
- **DAC:** ADV7123 (on-board, connected to VGA port).
- **Components:** Breadboard, 1kΩ pull-up resistor, 10kΩ resistors for voltage dividers.
- **Instrumentation:** Rigol DS1054Z Oscilloscope, Waveform Generator, Power Supply.

Software:

- Intel Quartus Prime (for VHDL synthesis).
- VHDL entities provided: **sar_full**, **comparator_test**, **dac_test**.

Experimental Setup: The VGA Red channel (DAC output) was connected to the inverting input (V_-) of the comparator. The signal to be converted (V_{in}) was applied to the non-inverting input (V_+). The comparator output was routed back to the FPGA via a GPIO pin to drive the SAR Finite State Machine.

3. Procedure and Measurements

Project 1: Comparator Characterization We mounted the LM393 on a breadboard powered at +5V. We connected a 1k Ω pull-up resistor to the output. We applied a reference voltage of 2.5V to one input, while a 50 kHz square wave was applied to the other to measure propagation delay. We measured:

- **Rising Edge Delay (t_{PLH}):** Dependent on the pull-up resistor and load capacitance.
- **Falling Edge Delay (t_{PHL}):** Dependent on the open-collector transistor's driving capability. These values were needed for calculating the `clkdivisor` parameter for the SAR logic.

Project 2: DAC Characterization Using the `dac_test` entity, we validated the linearity of the ADV7123 (Red channel). We measured the output voltage across the 75 Ω load resistor for 32 different digital codes (focusing on the 5 MSBs). The theoretical maximum voltage range was calculated as: $V_{max} = 18.5\text{mA} * 75\Omega \approx 1.39\text{V}$

Project 3 & 4: Full SAR Implementation We loaded the `sar_full` entity. We configured the parameters via switches:

- **sar_nbit:** Set to 3 (implying 3+2=5 bits resolution) or higher.
- **clkdivisor:** Calculated to ensure the clock period T_{slow} exceeded the measured comparator delay.
- **Feedback Loop:** The comparator output was connected to GPIO_0(1).

We tested the ADC with:

- **DC Voltages:** Varying V_{in} from 0V to 1.3V and verifying the hex output on the 7-segment display.
- **Dynamic Signals:** Using a sinusoidal input ($f=500\text{Hz}$, amplitude within 0-1.4V range) and observing the tracking reconstruction on the Green VGA channel.

4. Results

Comparator Analysis: The LM393 characterization measured the dynamic response of the device. Using a square wave test signal (Figure 3 in Appendix B), we measured the switching times. As detailed in **Figure 2**, we observed a low-to-high transition (t_{PLH}) of approximately **290 ns** and a high-to-low transition (t_{PHL}) of approximately **330 ns**. These values align with the datasheet specifications for large signal response, confirming the open-collector output's dependence on the pull-up resistor for the rising edge.

System Operation: To verify the signal integrity before closing the loop, we generated a test signal. **Figure 3** confirms the test frequency of 100 kHz (period 10.0 μ s). In full SAR operation, the DAC output exhibited the characteristic "staircase" waveform, converging toward the input voltage V_{in} over N clock cycles (Successive Approximation steps).

Conversion Accuracy: The system's flexibility was verified by changing the resolution. **Figure 4** (Appendix B) shows the operation in 4-bit mode. When the resolution was increased to 5 bits (**Figure 5**), the system correctly performed an additional approximation step, successfully tracking the input voltage with finer granularity. When the input exceeded the DAC range ($>1.4V$), the output correctly saturated at 11111. We observed errors primarily when the clkdivisor was set too low, causing the logic to latch the comparator value before the analog signal had settled.

5. Discussion

The experiment highlighted the trade-off between speed and power in mixed-signal design. The use of a 1k Ω pull-up resistor on the LM393 provided a reasonable rising edge speed (1 μ s), but a lower resistance would have improved speed at the cost of higher current consumption. The effective sampling rate f_S of the SAR ADC is given by:

$$f_S = f_{clk_slow} / N$$

This relation reflects that a full conversion requires N clock cycles. Given the comparator delay constraint ($T_{clk_slow} > 1.5 \mu$ s), the maximum sampling rate for an 8-bit conversion is limited to roughly 60-80 kSps. This confirms that while the digital logic (FPGA) and DAC (240 MHz) are fast, the external analog comparator is the bottleneck of this specific implementation.

6. Conclusions

We successfully designed and characterized a SAR ADC. The lab demonstrated the practical challenges of interfacing high-speed digital logic (FPGA) with slower analog components. The measurements confirmed the theoretical operation of the binary search algorithm and emphasized the importance of timing analysis in feedback loops. The resulting ADC, while limited in speed by the LM393, provided a functional platform for understanding A/D conversion principles.

7. References

- Texas Instruments. *LM393 Datasheet*.
- Analog Devices. *ADV7123 Datasheet*.

Appendix B – Figures and Tables

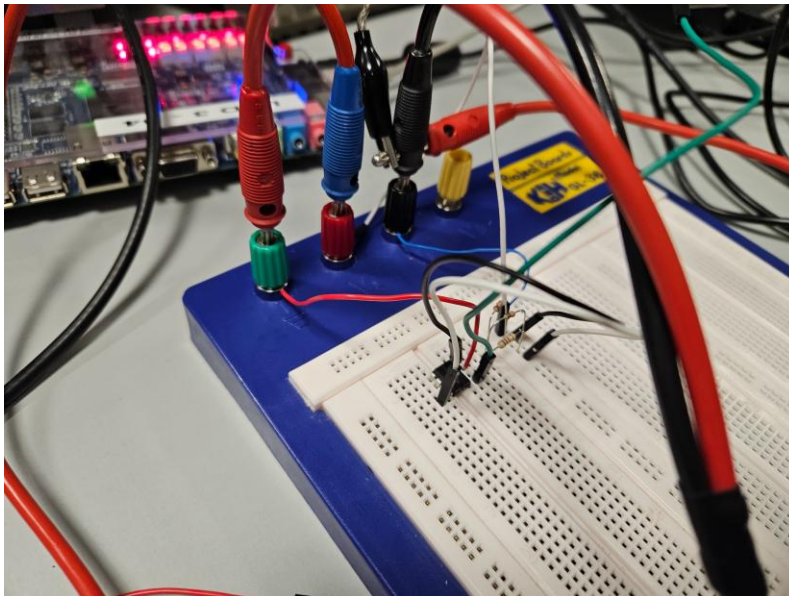


Figure 1: Close-up view of the analog circuit implementation on the breadboard. The LM393 comparator is visible in the center, configured with the required 1 kΩ pull-up resistor (visible near the jumper wires) to interface the open-collector output with the FPGA's digital logic. The wiring connects the comparator inputs to the DAC output and the input signal V_{in} .

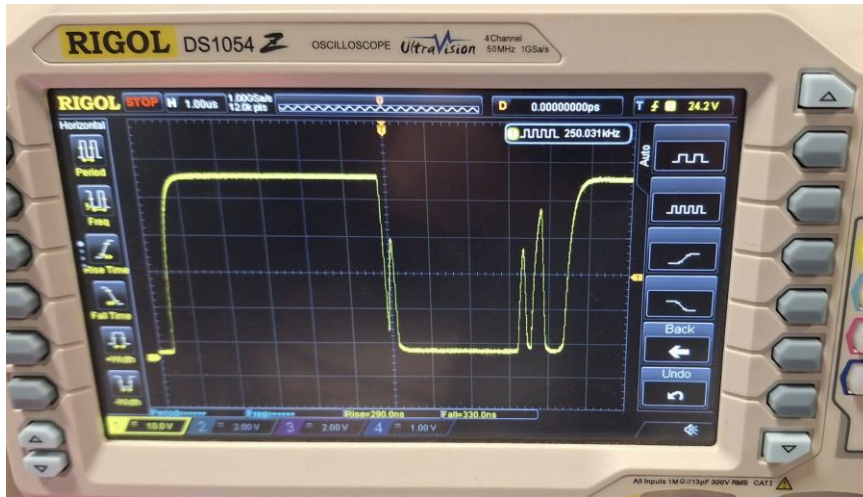


Figure 2: Oscilloscope capture showing the dynamic characterization of the LM393 comparator. The automatic measurements at the bottom of the display indicate a Rise Time (t_{PLH}) of 290.0 ns and a Fall Time (t_{PHL}) of 330.0 ns, confirming the timing constraints used to calculate the SAR clock divisor.

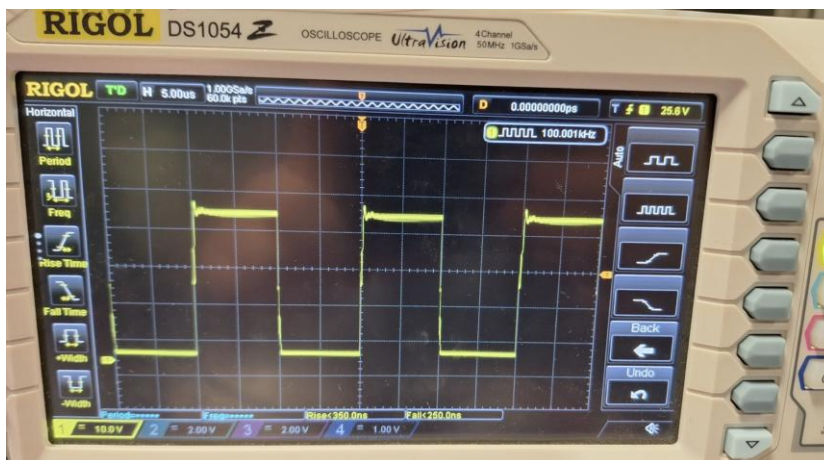


Figure 3: Oscilloscope capture of the square wave test signal generated for the initial comparator characterization (Project 1). The automatic measurement confirms a frequency of 100.001 kHz (period $T = 10.0 \mu$ s), validating the signal generator configuration used to stress the comparator's dynamic response.



Figure 4: Oscilloscope capture demonstrating the Successive Approximation algorithm in action. The yellow trace (CH1) monitors the DAC output (Red channel), displaying the characteristic "staircase" waveform as the logic tests each bit from MSB to LSB to converge on the input voltage V_{in} before resetting.



Figure 5: Oscilloscope capture of the SAR ADC operating in 5-bit configuration. Compared to the 4-bit test (Figure 4), the staircase waveform (Yellow trace, CH1) performs an additional approximation step to achieve higher resolution before the end-of-conversion reset.